

## **Assignment #10 - Final Design Reports**

### **Table of Contents for Fall Design Reports**

#### **Team Members and Project Abstract**

##### **Team Members:**

- Mitch Koski
- Muneer Al-Khasawneh
- Muhanad Al-Khasawneh

**Advisor:** Jeremy Hill

##### **Project Abstract:**

PrintNet is a centralized 3D printing lab management system designed to streamline the submission, tracking, and completion of print jobs. The platform improves workflow efficiency by organizing requests, managing printer resources, and providing clear communication between users and operators. We want PrintNet to utilize the cloud which is superior to current 3d printing systems currently in place. This report combines revised and updated work from all Fall assignments.

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#### **Project Description (Assignment #2)**

 GitHub Link 2: <https://github.com/Electrovian/Project-PrintNet/tree/Main/Classwork/assignment2>

##### **Summary:**

This section provides an overview of the PrintNet system, including the motivation for the project, target users, and the problems the platform solves within the 3D print lab environment.

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## User Stories and Design Diagrams (Assignment #4)

 GitHub Link 4: <https://github.com/Electrovian/Project-PrintNet/tree/Main/Classwork/assignment4>

### Summary:

The design diagrams show how the PrintNet system handles print requests.

Level 0 gives a simple overview of users interacting with the system.

Level 1 breaks the system into main parts such as job submission, the job queue, and printer management.

Level 2 shows more detailed steps like validating files, scheduling prints, and notifying users.

Boxes represent system components, and arrows show the direction of data flow. The diagrams help us understand how information moves through the system and how each part works together.

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## Project Tasks and Timeline (Assignments #5–6)

 GitHub Link 5: <https://github.com/Electrovian/Project-PrintNet/tree/Main/Classwork/assignment5>

 GitHub Link 6: <https://github.com/Electrovian/Project-PrintNet/tree/Main/Classwork/assignment6>

### Summary:

These assignments include our project task list, timeline, and workload distribution. They show the planned schedule for development and how responsibilities were divided among team members.

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## **ABET Concerns Essay (Assignment #7)**

 GitHub Link 7: <https://github.com/Electrovian/Project-PrintNet/tree/Main/Classwork/assignment7>

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## **PPT Slideshow (Assignment #8)**

 GitHub Link 8: <https://github.com/Electrovian/Project-PrintNet/tree/Main/Classwork/assignment8>  GitHub Link 8: <https://github.com/Electrovian/Project-PrintNet/tree/Main/Classwork/assignment8>

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## **Self-Assessment Essays (Assignment #3)**

 GitHub Link 3: <https://github.com/Electrovian/Project-PrintNet/tree/Main/Classwork/assignment3>

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## **Professional Biographies (Assignment #1)**

 GitHub Link 1: <https://github.com/Electrovian/Project-PrintNet/tree/Main/Classwork/assignment1>

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## **Budget**

No direct expenses were required for this project. We used personal computers, free software, and university resources.

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## Appendix

 Code Repository: <https://github.com/Electrovian/Project-PrintNet>

All team members met the required effort:

- Mitch Koski: **100+ hours(stopped counting)**
- Muneer Al-Khasawneh: **68+ hours**
- Muhanad Al-Khasawneh: **63+ hours**

Work included coding, research, documentation, meetings, and communication through Discord, in person, messaging/emailing and Teams.